

## REAL TIME GROSS SETTLEMENT (RTGS) - A PAYMENT SYSTEM

*Pradeep Naik\**

Payment systems are the means by which funds are transferred between banks. It constitutes channels by which risk can be minimised across domestic and international financial systems and markets. Robust payment systems are, therefore, a key requirement for maintaining and promoting financial stability.

The Payment System, which effects payment between a payer and a beneficiary, has importance for the functioning and integration of financial markets. It influences the speed, financial risk, reliability and cost of domestic and international transactions. The payment system also affects the transmission process in monetary management, the efficiency of financial intermediation, and the overall pace of financial reform.

As per Bank for International Settlements (BIS), Payment system operates on the following generally accepted service standards.

-Anyone can make payments to whomsoever one likes, whenever one likes, in whatever type of currency one likes, at the cost of a few cents per transaction. There are no settlement delays or mountains of

paperwork and value is received instantaneously. There are no distinctions in costs or delays between a domestic and a foreign currency transaction. Interest is computed real-time rather than on a “settlement day”, a relic from the ancient times, when accounting was done manually. Finally, privacy and security are guaranteed.-

Payment systems can be subject to a range of risks, including:

**Credit risk:** the risk that a party within the system will not be able to fully meet its financial obligations within the system currently or at any time in the future;

**Liquidity risk:** the risk that a party within the system will have insufficient funds to meet financial obligations within the system, as and when expected, although it may be able to do so at some time in the future;

**Legal risk:** the risk that a poor legal framework or legal uncertainties will cause or exacerbate credit or liquidity risks;

**Operational risk:** the risk that operational factors such as technical malfunctions or operational mistakes will cause or exacerbate credit or liquidity risks; and

*\*Shri Pradeep Naik, is Assistant Vice-President, Funds Settlement & Margins Department,  
The Clearing Corporation of India Limited.*

**Systemic risk:** in the context of payment systems this is the risk that the inability of one of the participants to meet its obligations, or a disruption in the system itself, could result in the inability of other system participants or of financial institutions in other parts of the financial system to meet their obligations, as they become due. Such a failure could cause widespread liquidity or credit problems and, as a result, could threaten the stability of the system or of financial markets.

#### **Systemically Important Payment Systems in India:**

The Working Group on Systematically Important Payment Systems, set up by the RBI, has identified the following payment systems as systemically important based on, inter-alia; whether the participants are banks or not, whether the settlement takes place in the books of Central bank or not, whether the individual value of payments is typically small or not, and whether the transactions relate to financial markets or not, whether the failure of one or more participants has a cascading and contagious effect on other participants and other systems :

- The Inter-bank Clearing System;
- The High Value Clearing System;
- The Securities Clearing and Settlement System;
- The MICR Clearing System;
- The Government Securities and Foreign

Exchange Clearing Systems; and,

- The proposed Real Time Gross Settlement System.(RTGS)

The RTGS Solution is soon to be rolled out in the Indian financial environment by the Reserve Bank of India (RBI).

#### **RTGS Concepts and Working of the RTGS System**

The Inter-bank payment systems handle large amounts of money. Many of these payments are for the settlement of financial market transactions between the banks themselves, such as Inter-bank loans / foreign exchange deals / money market operations and other payments are made by banks on behalf of their customers.

An RTGS payment system is one in which payment instructions between banks are processed in an electronic mode and settled individually and continuously throughout the day. It is a real time settlement system which effects final settlement continuously rather than periodically at pre-specified times provided that a sending bank has sufficient balance cover or credit. The attraction of the RTGS systems is that payee banks and their customers receive funds with certainty and finality during the day enabling them to use the funds immediately without exposing themselves to risk. This is in contrast to net settlement systems (such as paper-based clearing houses and their more modern electronic equivalents), where payment instructions are processed

(“cleared”) throughout the day but Inter-bank settlement (i.e. the movement of the funds between the banks) takes place only afterwards, typically at the end of the day. An RTGS system can thus be characterized as a funds transfer system that is able to provide continuous intraday finality for individual transfers. In many cases the transfer orders are processed and settled with the extensions of central bank credit, normally provided for a period of less than one business day (intraday credit). Thus RTGS systems are able to minimize or even eliminate the basic inter-bank risks in the settlement process.

#### **Message Flow Structures:**

To initiate a funds transfer, the sending bank dispatches a payment message which is subsequently routed to the central bank and to the receiving bank as the system processes and settles the transfer. Arrangements for routing payment messages in the majority of RTGS systems are based on a so-called V-shaped message flow structure. In this structure the full message with all the information about the payment (including, for example, the details of the beneficiary) is initially passed to the central bank and is sent to the receiving bank only after the transfer has been settled by the central bank.

Some RTGS systems, and particularly those that use the S.W.I.F.T. network, apply an alternative structure, known as a Y-shaped structure. In this case, the payment message

is transmitted by the sending bank to a central processor. The central processor takes a subset of information that is necessary for settlement from the original message and routes this core subset to the central bank (the original message being kept in the central processor). Upon receipt of the core subset, the central bank checks that the sending bank has sufficient cover funds on its account and informs the central processor of the status of the transfer, for instance, queued or settled. Once settled, the full message containing the confirmation of settlement is rebuilt by the central processor and sent to the receiving bank. The business information exchanged between the sending and receiving bank is, therefore, not known by the settlement agent.

A third structure is L-shaped structure, which is conceptually similar to a Y-shaped structure. The payment message dispatched by the sending bank is held at a system “gateway” attached to the sending bank’s internal processing system and a subset of information contained in the original message (a settlement request) is sent to the central bank. If the sending bank has sufficient cover funds on its account, settlement is completed and the central bank sends back a confirmation message to the sending bank’s gateway. Only upon receipt of this confirmation, the original payment message is released automatically from the gateway of the sending bank and sent to the receiving bank.

These different structures reflect differences in both the network configuration of the system and the operational role of the central bank. In both the V and Y-shaped structures, all messages from the sending bank are routed first to a central entity (S.W.I.F.T., a central processor or the central bank itself) and after settlement, all messages to the receiving bank are sent by that central entity. By contrast, there is no central entity for delivering messages in the L-shaped structure, where message routing is based on the bilateral exchange of information between banks, reflecting the decentralized architecture of the CHAPS system. In terms of its operational role in the system, the central bank is directly involved in both the settlement and processing of payment messages in the V-shaped structure, while in the Y and L-shaped structures the process of message routing can be handled by the network operator or the banks themselves, with the central bank only acting as the settlement agent.

These above three types of structure share the common feature that the receiving bank will receive the full payment message only after the transaction has been settled by the central bank. In these structures, therefore, the message flow structure per se cannot give rise to the possibility that the receiving bank will act upon unsettled payments.

Alternatively, RTGS systems could apply a T-shaped structure, in which the sending

bank routes payment messages directly to the receiving bank (through the message carrier), with copies (made by the message carrier) sent simultaneously to the central bank. This means that the receiving bank will first receive the full, unsettled payment message immediately after the sending bank has dispatched it and will subsequently also receive a confirmation message from the central bank once settlement has taken place. The T-shaped structure has generally been viewed as being incompatible with the basic principle of RTGS that a funds transfer should be passed to a receiving bank, if and only if, it has been settled irrevocably and unconditionally by the central bank.

Reserve Bank of India has chosen Y – shaped structure to meet a strategic objective i.e. the option to hive-off the Inter-bank Funds Transfer Processor (IFTP), which strips and retains the customer-related information and forwards the payment and settlement particulars to the RTGS, to an independent industry service provider.

### **Queuing Arrangements:**

Queuing refers to an arrangement whereby funds transfer orders are held pending by the sending bank or by the system in a certain order so as to prevent any limits set against the sending bank from being breached or to manage liquidity more generally. In RTGS systems, queues are most commonly generated when sending banks do not have sufficient cover funds in

their central bank account. In our proposed RTGS system queues will be generated only when sending banks will not have sufficient cover funds in their settlement account with RBI.

Most centrally located queuing arrangements have adopted a form of FIFO (“first in, first out”) rule for queue processing. Where the FIFO rule is applied, funds transfer orders are held in the order in which they are dispatched by the sending bank; the payment at the top of the queue is released and settled when covering funds become available, and only then is the payment behind it in the queue considered for settlement. However, to enable the participants to take care of urgent or time critical payments and to enable more effective funds management, the system allows the participants to assign priorities to their payment messages and thereby, enabling a particular transaction to be processed before another transaction, which was submitted earlier to the RTGS system. Within payment messages having the same priority, however, the transactions will be processed on FIFO basis.

Payment transactions, emanating from a participant are ordinarily expected to be settled immediately after it is received. This is the essence of a real time system. However, the possibility of some transactions not being immediately settled cannot be completely ruled out for the want of sufficient funds in the senders’

settlement account with RBI. To meet such exigencies, the proposed RTGS system in India provides for the maintenance of participant-wise payment queues in which payment transactions will be held in FIFO order within priority pending settlement. The system also provides for facilities to the participants to view their respective transactions held in their payment queues, cancel such transactions and even change their priority. However, participants can only view transactions on their own payment queues. They cannot view other participants’ queues or their own pending incoming payment instructions.

The proposed RTGS system by RBI provides for Participant’s Dedicated Settlement Account for RTGS Transactions. Each participant of the RTGS system will be required to open a dedicated **RTGS Settlement Account** for putting through its RTGS transactions. This account will be an *intra-day* account in the sense that it would be operational only during the duration of the RTGS day. The account would be funded at the start of the day (SOD) from a current account, the participant holds under the present dispensation at DAD, Mumbai. Balances in the RTGS Settlement account at the End of Day (EOD) of the RTGS day are swept back to the Participant’s current account and thereby zeroing the RTGS settlement account.

The system enables the participants to place

standing instructions with DAD, Mumbai to fund their RTGS settlement account each morning. They can specify an actual amount or percentage of balance to be transferred to the RTGS settlement account every day at SOD. Participants can also specify a minimum threshold value, which must be maintained in their current account while funding their RTGS settlement account. The system also provides a facility to fund the RTGS settlement account during the day from the participants' current accounts by the use of Own Account Transfers

The system provides for the inter-bank transaction type, which can be used to settle participants' financial obligations on their own account. The solution also provides for a separate transaction type for customer payments, which can be used to transmit the customer information along with the payment message to the beneficiary's bank in a structured format, which enables Straight Through Processing (STP) at the receiving participant's end. Delivery Vs. Payment (DVP) transactions, emanating from the RBI's Securities Settlement System, will also be settled in the RTGS system as a separate transaction type.

To provide for further flexibility, the system enables the use of code words to define new transaction types, based on the aforesaid primary transaction types, to meet the specific needs of the participants.

The Multilateral Net Settlement batches

(the new avatar of the existing Clearing batches) will also be settled in the RTGS system. This will ensure more efficient settlement of net clearing batches including the settlement of the transactions, emanating from the Clearing Corporation of India Ltd. (CCIL) and other potential clearing and settlement agencies as also more efficient monitoring of un-cleared credit positions.

#### **Liquidity in RTGS systems:**

In general terms, for an individual participant in an RTGS system the three possible sources of funds are (1) balances maintained in the account with the central bank, (2) incoming transfers from other banks including borrowing through the money market and (3) credit extensions from the central bank.

RTGS systems entail active management of intraday liquidity. Adequate liquidity, relative to the value and distribution of payments, makes a smooth flow of payments possible through such systems, helping to avoid delays to individual payments and minimising liquidity risks. It is imperative that participants ensure at the time of submission of payment instructions, that there are sufficient funds in their RTGS settlement account to settle their transactions.

The proposed RTGS system provides several facilities to supplement the participants' liquidity management efforts. The

provision of a separate RTGS settlement account, own account transfers, queuing facilities and priority assignment are all such tools. Other powerful tools to be provided by RBI are:

- 1. Intra Day Liquidity:** Intra-day liquidity will be provided by RBI at its discretion and under terms and conditions, to be specified from time to time. Such intra-day liquidity will be fully collateralized and will be provided to the participants at a charge per transaction. However, failure to repay the credit before the end of the day will invite strict penal action.
- 2. Gridlock Resolution Mechanism:** Gridlock is a situation where payment

of one participant can not be processed unless and until the receipt of the funds from the other participant and vice versa. In RTGS system there will be a possibility of potential grid locks which may arise when two or more participants have receipts and payments for the like amounts among themselves. Such crippling liquidity problems can be overcome by this mechanism. RBI may invoke Gridlock Resolution Mechanism periodically to enable smooth settlement of the RTGS transactions.

Thus RTGS system is a robust payment system which effects final settlement continuously with certainty and finality.